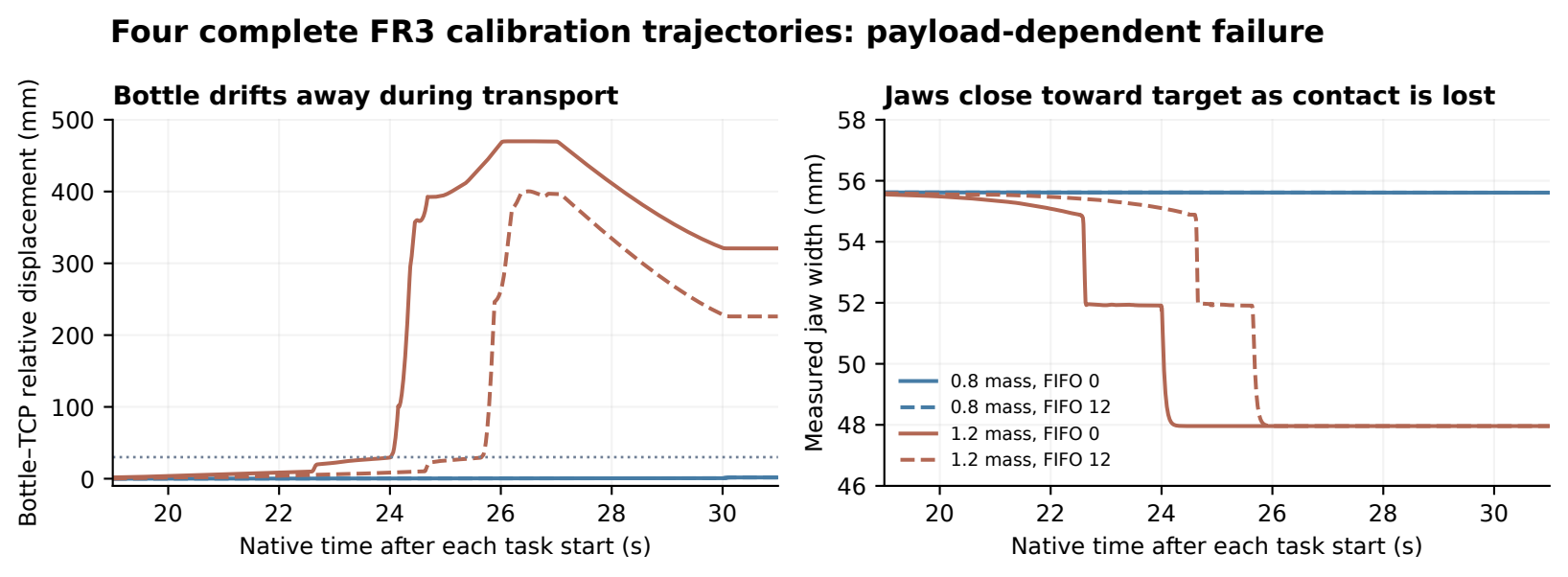
Four complete FR3 calibration trajectories: payload-dependent failure
Bottle drifts away during transport
Bottle–TCP relative displacement (mm)
0
100
200
300
400
500
20
22
24
26
28
30
Native time after each task start (s)
Jaws close toward target as contact is lost
Measured jaw width (mm)
46
48
50
52
54
56
58
0.8 mass, FIFO 0
0.8 mass, FIFO 12
1.2 mass, FIFO 0
1.2 mass, FIFO 12
20
22
24
26
28
30
Native time after each task start (s)